COLLEGE OF ENGINEERING, GUINDY

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

MINI PROJECT ABSTRACT

SUBJECT: DATA STRUCTURES USING C
TOPIC: CANTEEN ORDER MANAGEMENT SYSTEM

SUBMITTED BY:

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CANTEEN ORDER MANAGEMENT SYSTEM USING C

1. Explanation

The Canteen Order Management System is a Data Structures mini project developed using the C programming language to simplify food ordering and order management in a college canteen. The program displays the available food items and their prices, accepts the selected food items and quantities, and calculates the total bill. Each order is assigned a token number for easy identification. The system uses fundamental data structures to store food details, manage pending orders, and process orders in the order they are received. This project demonstrates the practical application of data structures in solving everyday problems.

2. Data Structures and Concepts Used

- Structures (struct)
- Arrays
- Circular Queue
- Searching
- String Handling
- Functions
- Control Statements

3. Data Structures and Their Purpose

Structures (struct): Structures store related information such as food ID, food name, price, student name, token number, and total bill.

Arrays: Arrays store the available food items and order details for easy access and management.

Circular Queue: A circular queue manages pending orders using the FIFO (First In, First Out) principle, ensuring that the earliest order is processed first.

Searching: Searching helps locate food items using their IDs and find pending orders using their token numbers.

String Handling: String handling manages text information such as student names and food item names.

Functions: Functions divide the program into separate operations, including displaying the menu, placing orders, calculating bills, generating tokens, and processing orders.

Control Statements: Control statements manage the flow of the program using decision-making and looping statements.

4. Main Modules

- **Menu Display:** Displays the available food items and their prices.
- **Order Placement:** Accepts the student's name, selected food items, and quantities.
- **Bill Calculation:** Calculates the total cost based on the selected items and quantities.
- **Token Generation:** Assigns a token number to each order.
- **Queue Management:** Stores pending orders and processes them in FIFO order.
- **Order Searching:** Searches for food items and pending orders using their IDs or token numbers.

5. Real-World Applications

- Helps organise food ordering in college canteens and small food counters.
- Reduces confusion when multiple students place orders during break times.
- Provides a basic token-based ordering system.
- Helps manage food items, quantities, and billing systematically.
- Demonstrates the use of queues in real-world order-processing systems.

6. Advantages

- Simple and easy-to-understand C program.
- Reduces manual effort in calculating bills.
- Organises orders using the FIFO principle.
- Makes order identification easier through token numbers.
- Demonstrates important Data Structures concepts using C.
- Improves understanding of data structures through a practical application.

7. Conclusion

The Canteen Order Management System demonstrates how fundamental data structures can be used to solve a practical problem in a college environment using the C programming language. The program supports menu display, order placement, token generation, bill calculation, searching, and queue-based order processing. By implementing structures, arrays, and circular queues, this project provides a practical understanding of data structures and their applications in managing canteen orders efficiently.